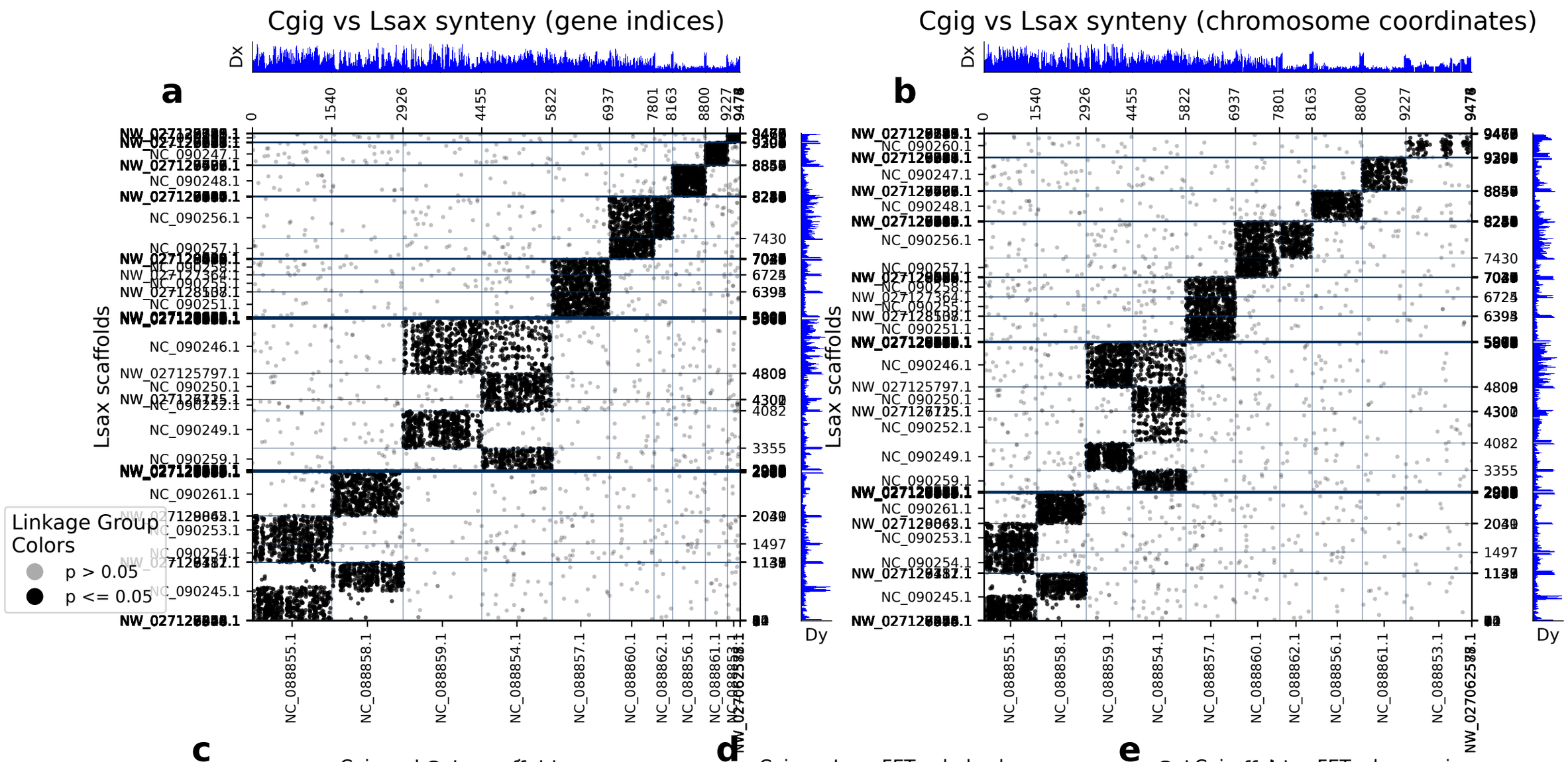
diamond: 9479 orthologs on 13 Cgig scaffolds and 205 Lsax scaffolds.



c

Cgig vs Lsax synteny (gene indices)

Cgig scaffolds

d

Cgig vs Lsax FET, whole chroms.

	Cgig scaf	Lsax scaf	p (FET)	Count
0	NC_088858.1	NC_090261.1	0.00e+00	767
1	NC_088859.1	NC_090246.1	0.00e+00	735
2	NC_088859.1	NC_090249.1	0.00e+00	651
3	NC_088855.1	NC_090245.1	9.64e-186	571
4	NC_088856.1	NC_090248.1	0.00e+00	514
5	NC_088858.1	NC_090245.1	2.44e-124	468
6	NC_088855.1	NC_090253.1	6.80e-301	464
7	NC_088854.1	NC_090250.1	0.00e+00	445
8	NC_088857.1	NC_090251.1	0.00e+00	431
9	NC_088860.1	NC_090256.1	9.05e-243	413
10	NC_088854.1	NC_090259.1	1.58e-242	356
11	NC_088861.1	NC_090247.1	0.00e+00	345
12	NC_088860.1	NC_090257.1	1.57e-297	326
13	NC_088862.1	NC_090256.1	4.04e-273	299
14	NC_088855.1	NC_090254.1	1.47e-169	291
15	NC_088857.1	NC_090255.1	7.94e-221	283
16	NC_088857.1	NC_090258.1	6.65e-167	239
17	NC_088854.1	NC_090246.1	6.71e-07	222
18	NC_088854.1	NC_090252.1	6.12e-78	154
19	NC_088853.1	NC_090260.1	1.22e-164	120

e

Cgig vs Lsax FET, chrom. pieces

	Cgig scaf	Lsax scaf	p (FET)	Count
0	NC_088858.1 : 0-end	NC_090261.1 : 0-end	0.00e+00	767
1	NC_088859.1 : 0-end	NC_090246.1 : 0-end	0.00e+00	735
2	NC_088859.1 : 0-end	NC_090249.1 : 0-end	0.00e+00	651
3	NC_088855.1 : 0-end	NC_090245.1 : 0-end	9.64e-186	571
4	NC_088856.1 : 0-end	NC_090248.1 : 0-end	0.00e+00	514
5	NC_088858.1 : 0-end	NC_090245.1 : 0-end	2.44e-124	468
6	NC_088855.1 : 0-end	NC_090253.1 : 0-end	6.80e-301	464
7	NC_088854.1 : 0-end	NC_090250.1 : 0-end	0.00e+00	445
8	NC_088857.1 : 0-end	NC_090251.1 : 0-end	0.00e+00	431
9	NC_088860.1 : 0-end	NC_090256.1 : 0-end	9.05e-243	413
10	NC_088854.1 : 0-end	NC_090259.1 : 0-end	1.58e-242	356
11	NC_088861.1 : 0-end	NC_090247.1 : 0-end	0.00e+00	345
12	NC_088860.1 : 0-end	NC_090257.1 : 0-end	1.57e-297	326
13	NC_088862.1 : 0-end	NC_090256.1 : 0-end	4.04e-273	299
14	NC_088855.1 : 0-end	NC_090254.1 : 0-end	1.47e-169	291
15	NC_088857.1 : 0-end	NC_090255.1 : 0-end	7.94e-221	283
16	NC_088857.1 : 0-end	NC_090258.1 : 0-end	6.65e-167	239
17	NC_088854.1 : 0-end	NC_090246.1 : 0-end	6.71e-07	222
18	NC_088854.1 : 0-end	NC_090252.1 : 0-end	6.12e-78	154
19	NC_088853.1 : 0-end	NC_090260.1 : 0-end	1.22e-164	120

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Lsax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 9479 orthologs found between 13 Cgig scaffolds and 205 Lsax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.